PREMIS Data Dictionary for Preservation Metadata

Version 4 DRAFT

PREMIS Editorial Committee

2026

http://www.loc.gov/standards/premis

CONTENTS

INTRODUCTION

The PREMIS Data Dictionary is a comprehensive, practical resource for implementing preservation metadata in digital preservation systems. The Data Dictionary defines preservation metadata that:

- Supports the viability, renderability, understandability, authenticity, and identity of digital objects in a preservation context;
- Represents the information most preservation repositories need to know to preserve digital materials over the long- term;
- Emphasizes "implementable metadata": rigorously defined, supported by guidelines for creation, management, and use, and oriented toward automated workflows; and
- Embodies technical neutrality: no assumptions made about preservation technologies, strategies, metadata storage and management, etc.

Background

Development of the Original PREMIS Data Dictionary

In June 2003, OCLC and RLG jointly sponsored the formation of the PREMIS (***Pre****servation* ***M****etadata:* ***I****mplementation* ***S****trategies*) working group, comprised of international experts in the use of metadata to support digital preservation activities. The working group's membership included more than 30 participants, representing five different countries and a variety of domains, including libraries, museums, archives, government agencies, and the private sector. Part of the working group's charge was to develop a core set of implementable preservation metadata, broadly applicable across a wide range of digital preservation contexts and supported by guidelines and recommendations for creation, management, and use. This portion of the working group's charge was fulfilled in May 2005 with the release of *Data Dictionary for Preservation Metadata: Final Report of the PREMIS Working Group.* In addition to the Data Dictionary, the working group also published a set of XML schemas to support implementation of the Data Dictionary in digital preservation systems.

The PREMIS working group was established to build on the earlier work of another initiative sponsored by OCLC and RLG: the Preservation Metadata Framework (PMF) working group. In 2001–2002 the PMF working group outlined the types of information that should be associated with an archived digital object. Their report, *A Metadata Framework to Support the Preservation of Digital Objects* (the *Framework*), proposed a list of prototype metadata elements.[1] However, additional work was needed to make these prototype elements implementable. The PREMIS

[1] *A Metadata Framework to Support the Preservation of Digital Objects* (Dublin, Ohio: OCLC Online Computer Library Center, 2002), http://www.oclc.org/research/projects/pmwg/pm_framework.pdf.

working group was asked to take the PMF group's work a step further and develop a data dictionary of core metadata for archived digital objects, as well as give guidance and suggest best practice for creating, managing, and using the metadata in preservation systems.

Since the PREMIS working group had a practical rather than theoretical focus, members were sought from institutions known to be operating or developing preservation repository systems within the cultural heritage and information industry sectors. Diverse perspectives were also sought. The working group consisted of representatives from academic and national libraries, museums, archives, government, and commercial enterprises in five different countries. In addition, PREMIS called upon an international advisory committee of experts to review progress.

To understand how preservation repositories were actually implementing preservation metadata, in November 2003 the working group undertook a survey of about 70 organizations thought to be active in or interested in digital preservation, resulting in the report *Implementing Preservation Repositories for Digital Materials: Current Practice and Emerging Trends in the Cultural Heritage Community* (the *Implementation Survey Report*).[2] The findings of this survey were extremely helpful in informing the working group's discussions as it developed the Data Dictionary.

Both the earlier *Framework* and the PREMIS Data Dictionary build on the Open Archival Information System (OAIS) reference model (ISO 14721).[3] The OAIS information model provides a conceptual foundation in the form of a taxonomy of information objects and packages for archived objects, and the structure of their associated metadata. The *Framework* can be viewed as an elaboration of the OAIS information model, explicated through the mapping of preservation metadata to that conceptual structure. The PREMIS Data Dictionary can be viewed as a translation of the *Framework* into a set of implementable semantic units. However, it should be noted that the Data Dictionary and OAIS occasionally differ in terminology usage; these differences are noted in the Glossary that accompanies this report. Differences usually reflect the fact that PREMIS semantic units require more specificity than the OAIS definitions provide, which is to be expected when moving from a conceptual framework to an implementation. Additionally, starting with version 3.0 the PREMIS Data Dictionary explicitly expanded its scope beyond repository boundaries in order to accommodate seamless metadata representation across the digital object life-cycle.

Implementable, core preservation metadata

The PREMIS Data Dictionary defines "preservation metadata" as *the information a repository uses to support the digital preservation process.* Specifically, the group looked at metadata supporting the functions of maintaining viability, renderability, understandability, authenticity, and identity in a preservation context. Preservation metadata thus spans a number of the

---

[2] *Implementing Preservation Repositories for Digital Materials: Current Practice and Emerging Trends in the Cultural Heritage Community* (Dublin, Ohio: OCLC Online Computer Library Center, 2004), http://www.oclc.org/research/projects/pmwg/surveyreport.pdf.

[3] *ISO 14721:2012: Space Data and information transfer system—Open archival information system (OAIS)—Reference model (OAIS)* (Geneva, Switzerland: International Organization for Standardization, Aug. 2012).

categories typically used to differentiate types of metadata: administrative (including Rights, obligations and permissions), technical, and structural. Particular attention was paid to the documentation of digital provenance (the history of an object) and to the documentation of relationships, especially relationships among different objects within the preservation repository. ~~Version 3.0 expanded the scope beyond repository boundaries in order to accommodate metadata representations across the object life-cycle.~~

The group considered a number of definitions of "core." In one view, core describes any metadata absolutely required under any circumstances. In another, core means that metadata is applicable to any type of repository implementing any type of preservation strategy. PREMIS uses this practical definition: *things that most working preservation repositories are likely to need to know in order to support digital preservation.* The words "most" and "likely" were chosen deliberately. Core does not necessarily mean mandatory, and some semantic units were designated as optional when exceptional cases were apparent.

The concept of "implementability" also required definition. Most preservation repositories deal with large quantities of data. Therefore, a key factor in the implementability of preservation metadata is whether the values can be automatically supplied and automatically processed by the repository. Whenever possible the group defined semantic units that do not require human intervention to supply or analyze. For example, controlled values from an authority list are preferred over textual descriptions.

The working group decided that the Data Dictionary should be wholly implementation independent. That is, the core metadata defines information that a repository needs to know, regardless of how, or even whether, that information is stored. For instance, for a given identifier to be usable, it is necessary to know the identifier scheme and the namespace in which it is unique. If a particular repository uses only one type of identifier, the repository would not need to record the scheme in association with each object. The repository would, however, need to know this information and to be able to supply it when exchanging metadata with other repositories. Because of the emphasis on the need to know rather than the need to record or represent in any particular way, the group preferred to use the term "semantic unit" rather than "metadata element." The Data Dictionary names and describes semantic units.

PREMIS Maintenance Activity

~~Following the release of the Data Dictionary in 2005, the PREMIS working group retired and the PREMIS Maintenance Activity, sponsored by the Library of Congress, was initiated to maintain the Data Dictionary and coordinate other work to advance understanding of preservation metadata and related topics. In addition to providing a permanent Web home for the Data Dictionary, XML schema, and related materials, the Maintenance Activity also operates the PREMIS Implementers Group (PIG) discussion list and wiki, conducts tutorials on the Data Dictionary and its use, and commissions focused studies on preservation metadata topics. The Maintenance Activity also established an~~ The PREMIS Editorial Committee is responsible for ~~further~~ ongoing maintenance and development of the Data Dictionary and the XML schema and promoting their use. The membership of the Editorial Committee reflects a variety of countries and institutional backgrounds. Other key maintenance activities include: providing a permanent Web home for the Data Dictionary, XML schema, and related materials, operating the PREMIS

Implementers Group (PIG) discussion list and wiki, conducting tutorials on the Data Dictionary and its use, as well as commissioning focused studies on preservation metadata topics.

Users identify errors, and provide feedback on ways that the Data Dictionary could be improved to increase its value and ease of application through a variety of mechanisms: Discussion of issues takes place on the PREMIS Implementers Group (PIG) discussion list and wiki, user group meetings take place at PREMIS Implementation Fairs that are organized in conjunction with major conferences and are advertised on the PREMIS website and through mailing lists, and requests for changes can be submitted as issues via the PREMIS GitHub repository (URL). ~~through the *PREMIS Data Dictionary and Schema Revision Process* specified at http://www.loc.gov/standards/premis/revision-process.html.~~

The members of the Editorial Committee revise the Data Dictionary when a sufficient level of commentary has accumulated to warrant doing so, making every effort to engage stakeholders in the process of revision. The Committee keeps the preservation community informed of issues being discussed, solicits comment on proposed revisions, and consults outside experts where appropriate.

Version History

Version	**Release Date**
Version 1.0	May 2005
Schema Version 1.0	*17 May 2005*
Schema Version 1.1	*27 September 2005*
Version 2.0	March 2008
Schema Version 2.0	*17 July 2008*
Version 2.1	January 2011
Schema Version 2.1	*6 January 2011*
Version 2.2	July 2012
Schema Version 2.2	*15 May 2012*
Schema Version 2.3	*4 August 2014*
Version 3	June 2015

Version 4; feb? 2026

~~Subsequent~~ Versions of the Dictionary have taken advantage of the increase in use of PREMIS and experience of implementing preservation solutions. They have included corrections of errors, clarifications of some semantic units, changes for consistency, and the addition of a few semantic units that resulted from requests to the PREMIS Editorial Committee. Version 2.0 was a major revision and version 2.1 added additional functionality, particularly in the Rights entity. Version 2.2 was considered non-substantial as there were no major changes affecting existing PREMIS descriptions. Version 2.3 contained only changes that affected the XML schema (by adding the ability to designate use of specific controlled vocabularies); these were not reflected in the Data Dictionary. ~~Further information about this mechanism is given under the section "Supplying Metadata Values".~~

Starting from version 2.2, a PREMIS OWL ontology is available alongside the XML Schema. It allows one to provide a PREMIS-endorsed flavor of the Data Dictionary so that one can express preservation metadata in RDF. This ontology does not replace but complements XML in areas where RDF may be better suited, such as querying or publishing preservation metadata, or connecting repository-specific data to externally maintained registries.

Version 3 of the Data Dictionary was a major revision again, that apart from adding some new semantic units, included several major changes: This version of the Data Dictionary, version 3, includes some major changes and additions to the Dictionary. These can be highlighted as:

- Repositioning of Intellectual Entity as a category of Object to enable additional description within PREMIS and linking to related PREMIS entities.
- Repositioning of Environments (i.e. hardware and software needed to use digital objects) so that they can be described and preserved reusing the Object entity. They can be described as Intellectual Entities and preserved as Representation, File or Bitstream Objects.
- Addition of physical Objects to the scope of PREMIS so that they can be described and related to digital objects.
- Addition of a new semantic unit to the Object entity: preservationLevelType to indicate the type of preservation functions expected to be applied to the object for the given preservation level.
- Addition of a new semantic unit to the Agent entity to express the version of software Agents: agentVersion (O, NR).
- Addition of a new semantic unit to the Event entity: eventDetailInformation.(O,R)
- Addition of the semantic unit authority, which provides documentation of the mechanism to indicate the use of a controlled vocabulary for semantic units whose data constraint recommends it

~~Major additions are discussed in detail below (see the “PREMIS Data Model” and “Environment” sections). Other additions are explained within the relevant section of the Dictionary.~~

Version 4 is another major conceptual and structural update, focusing primarily on the Rights Entity:

- Rephrasing the semantic container rightsGranted to rule, in order to broaden the application of the entity; so that it can record obligations in addition to permissions. Together with the addition of a ruleType to address either ‘permission’ or ‘obligation’.
- Addition of an extension for actions, to support machine readable content, grouped in a container unit together with actiontype (formerly known as ‘act’).
- Repositioning termOfGrant and termOfRestriction to a container of the linkingObjectIdentifier, renaming them to ‘applicableDatesOfRule’ and

'applicableDatesOfRestriction', allowing to register different dates for (subsets of) objects that are linked to this rule. This repositioning implies that the linkingRightsIdentifier of the Object is altered concurrently.

- Addition of a new semantic unit relatedEvent to the linking semantic units (both directions), to support the option to record an Event for a specific Object - Rights combination, for instance a reviewing event or a trigger event.

- Addition of a container unit that holds 'trigger' and 'duration' for a rule or restriction. Giving the opportunity to define what type of event (like 'first publication of an object' or a specific external event agreed with the rightsholder) will activate a rule/restriction and for how long (duration).

- Restructuring of the Rights Entity as a whole to be able to refer to the identifier regardless of the use of the extension, positioning the extension as an alternative for the rightsStatement, and merging the containers for separate rightsbases into one container called rightsInformation.

PREMIS Awards and Recognition

The PREMIS Data Dictionary was awarded the 2005 Digital Preservation Award, given under the auspices of the British Conservation Awards, the 2006 Society of American Archivists Preservation Publication Award, and was a finalist for the 2012 Digital Preservation Award for the most outstanding contribution to digital preservation in the last decade. In 2022, the PREMIS EC was awarded the DPC 20th Anniversary Award, recognizing the PREMIS Data Dictionary and the work of coordinating efforts across the community.

The PREMIS Data Model

The PREMIS Data Dictionary defines **semantic units**. Each semantic unit defined in the Data Dictionary is mapped to an entity that is organized within a simple data model. A semantic unit can therefore be understood as a property of an entity. The model defines four entities important in regard to digital preservation activities: Objects, Events, Agents and Rights[4]. 4 Figure 1 provides a graphical illustration of the PREMIS Data Model.

Figure 1

>> Rights Statement >> "Right"; "Assertion of right, obligation or permission"

>> Arrow from Right to itself

In Figure 1, entities are represented by boxes; relationships between entities are represented by arrows. When arrows are bi-directional, then ~~each~~ both entity types contain~~s~~ a semantic unit

[4] Other preservation metadata initiatives have developed other models. The National Library of New Zealand defines four types of entity: objects, files, processes, and metadata modification. Metadata Standards Framework—Preservation Metadata (Revised) (Wellington: National Library of New Zealand, June 2003), https://ndhadeliver.natlib.govt.nz/delivery/DeliveryManagerServlet?dps_pid=IE753269

allowing it to link to the other. So, for example, the Rights entity includes a semantic unit recording information about the relationship with an Agent, and the Agent entity includes a semantic unit recording information about associated Rights.

The arrow pointing from the Objects entity back to itself indicates that the semantic units defined in the Data Dictionary support the recording of relationships between Objects. ~~No other entity in the data model supports relationships of this type;~~ The same holds for Rights; in other words, while Objects can be related to other Objects, and Rights can be related to other Rights, Events cannot be related to other Events, and Agents cannot be related to other Agents, ~~and so on~~.

The entities in the PREMIS data model are defined as follows:

Object (or Digital Object): a discrete unit of information subject to digital preservation[5]. ~~Version 3 introduces the notion that~~ This can also be an environment used as part of the preservation process.

Environment: Technology (software or hardware) supporting a Digital Object in some way (e.g. rendering or execution). Environments can be described as Intellectual Entities and captured and preserved in the preservation repository as Representations, Files and/or Bitstreams.

Event: an action that involves or affects at least one Object or Agent associated with or known by the preservation repository.

Agent: person, organization, or software program/system associated with Events in the life of an Object, or with Rights attached to an Object. It can also be related to an environment Object that acts as an Agent.

Rights ~~Statement~~: assertion of one or more Rights, obligations, or permissions pertaining to an Object and/or Agent.

Each semantic unit defined in the Data Dictionary is mapped to one of the entities in the data model. In this sense, a semantic unit may be viewed as a property of an entity. For example, the semantic unit size is a property of an Object entity. Semantic units have values: for a particular object the value of size might be "843200004."

In most cases, a particular semantic unit is unambiguously a property of only one type of entity. The size of an object is clearly a property of the Object entity. In some cases, however, a semantic unit applies equally to two or more types of entity. For example, Events have outcomes, which could be a property of the Event or the Object or Agent affected. If a migration event creates a file that has lost some important feature, the loss of that feature might be considered an outcome of the event, and therefore a property of the Event entity. Alternatively, it might be considered an attribute of the new file, and therefore a property of the Object entity. When a semantic unit applies equally to multiple entity types, the decision has been taken that the

[5] Note that the PREMIS definition of an Object entity differs from the definition of digital object commonly used in the digital library community, which holds a digital object to be a combination of identifier, metadata, and data. This is not intended to be a conflict. The Object entity in our model is an abstraction defined only to cluster attributes (semantic units) and clarify relationships.

semantic unit should be associated with only one type of entity in the Data Dictionary. The data model relies upon links between the different entities to make these relationships clear. In the example above, the loss of the feature is treated as a detailed outcome of the Event, where the Event contains the identifier of the Object involved. What is important is that this association is arbitrary and is not meant to imply that a particular implementation is required. The choice of semantic unit is down to individual implementations.

In some cases a semantic unit takes the form of a **container** that groups a set of related semantic units. For example, the semantic unit identifier groups the two semantic units identifierType and identifierValue. The grouped subunits are called **semantic components** of the container. Some containers are defined as **extension containers**, to allow the use of metadata encoded according to an external schema. This enables PREMIS to be extended with metadata elements that are more granular, non-core, or otherwise out of scope for the Data Dictionary.

A **relationship** is a statement of association between instances of entities. “Relationship” can be interpreted broadly or narrowly, and expressed in many different ways. For example, the statement “Object A is of format B” could be considered a relationship between A and B. The PREMIS model, however, treats format B as a property of Object A. PREMIS reserves “relationship” for associations between two or more Object entities, Rights entities, or between entities of different types, such as an Object and an Agent.

In some cases a relationship defines an association between a specific combination of entities and another entity. Version 4 introduces this type of relationship for Events that take place with regard to one or more Objects in combination with one or more Rights.

More on Objects

Intellectual entities

An **intellectual entity** is a distinct intellectual or artistic creation that is considered relevant to a designated community in the context of digital preservation. An intellectual entity can include other intellectual entities and it may have one or more representations.

~~Before versions 3.0 of the Data Dictionary At one point Intellectual Entities were considered out of scope for PREMIS, since m~~Many types of intellectual entities - such as books, articles, images, archival collections, or statistical studies – do not necessarily need to be described as part of the preservation metadata and are well-served by descriptive metadata that supports their discovery and retrieval from outside the preservation metadata. Local implementations may model Intellectual Entities outside PREMIS or within it.

~~In version 3.0, one now has the choice of modeling Intellectual Entities outside PREMIS or within it PREMIS.~~

…

Representations

P4, last line: "Alternatively, the repository could choose to duplicate the SGML file and store two identical copies of it."

Files, Bitstreams, and filestreams

Complete Examples

Environments

P2, line 1: "~~Starting with PREMIS version 3.0,~~ Object entities can be used to capture Environments that are relevant to the content of the repository."

More on Events

P3, line 1: Up to version 3.0, in the data model objects are associated with events in two ways.

Add after P7 "A given object can be associated in these two ways with any number of events.": Version 4.0 introduces a third way to associate an object with events via the event identifier that is recorded in the linkingObject container of a Rightstatement, as the semantic component relatedEventIdentifier

Review Event

This is an Event that represents a review of the Rights status. Semantically, there are two types of Rights Review Events:

1. A review of the content of the Object, such as a privacy review, a review of Classified material, or of previously unreviewed objects. This request may be triggered by a change in statute, the need for a preservation action, or request for dissemination of a Classified, restricted, or unreviewed Object. Note: even the right for staff to take preservation actions may be limited with restricted content, to either individual named staff or a class/group of staff, such as staff with a security clearance or members of an indigenous or tribal group.
2. A review of the rightsStatement itself related to Copyright or Licensing status, which may be triggered by a request for dissemination or another external trigger, such as a change in statute or the passage of time, which then requires a review of the current rightsStatement.

Such review Events will result in an eventOutcome: no change, a partial update to Rights, or the addition of a new rightsStatement.

For review Events for classified or restricted Objects that result in a new redacted version of a file, the eventOutcome should include an appropriate Redaction Code, additional metadata that identifies the statute(s) under which the information remains redacted. It is a local implementation decision to treat a newly created version, such as a redacted version, as a new PREMIS object with its own rightsStatement.

A rightsStatement may also be updated instead of creating a new rightsStatement in cases where the outcome of the review Event is a wholesale change, such as the expiration of a restriction, license, or copyright. The Review Event as referred to in the linkingEventIdentifier is used to record and track the event leading to the change. Review events with partial outcomes (such as a partial declassification that results in redacted availability) must have a new rightsStatement.

Rights review Events will stack over time to provide a review history.

Trigger Event

Another example for the use of the linkingObject container of the Rightsstatement is when the related Event represents a trigger that activates a rule for a (set of) objects. The Event will have an eventDateTime, that corresponds with the *trigger* as described in the rule, namely what sort of date is used to activate the rule: Birthdate, Death Date, Publication Date, Review for Release, Declassification, etc.

More on Agents

Agents are clearly important but are not the focus of the Data Dictionary, which defines only a means to identify the agent and a classification of agent type (person, organization, or software). While more metadata is likely to be necessary, this is left to other initiatives to define in detail. In version 3.0, the agentVersion semantic unit is added to the Data Dictionary.

The data model diagram shows how Agents relate to other entities. An Agent can be related to Rights Statements in which the agent has an interest. For example, an Agent may be the source of the right (as the rightsholder or contract partner) or a resource (as the software that must be used). Indeed, the ‘target’ of a Rights Statement (to whom the rule applies) will not always be implicitly the repository. One example where Agents can be the recipient/target of a Rights Statement include a class of repository staff with the appropriate security clearances given the right to perform preservation actions. Another example is an indigenous/tribal group being given rights to access Objects that are not available to others.

An Agent can also relate to the Event entity in which the Agent takes an action. Each Event can have one or more related Agents. Because a single Agent can perform different roles in different Events, the role of the Agent is a property of the Event entity, not of the Agent entity. Agents relate to Objects in 2 ways: 1) Agents affect Objects that are involved in an Event; 2) Agents are software Agents that are described through an environment Object. In the first case Agents influence Objects only indirectly through Events and are not directly linked to the Object. In the second case the environment Object further describes and captures the software Agent. For this situation there is a direct link (linkingEnvironmentIdentifier) from the Agent to the environment.

More on Rights

Many efforts are concerned with metadata related to intellectual property rights and permissions, from rights expression languages to the <indecs[11]> framework. However, only a small body of work addresses rights and permissions specifically related to digital preservation. After the publication of the first edition of the PREMIS Data Dictionary, the Library of Congress in its capacity as PREMIS Maintenance Agency commissioned a paper, "Rights in the PREMIS Data Model," by Karen Coyle[12]. This paper discussed copyright, licenses, and statute as three bases for establishing intellectual property rights, and recommended an expansion of the rights information in the Data Dictionary to include information on these bases.

Consequently, the permissionStatement in the original Data Dictionary was replaced with the rightsStatement in version 2.0. In that revision the Editorial Committee relied heavily upon the Coyle paper, background materials such as Peter Hirtle's "Digital Preservation and Copyright,13" and the California Digital Library's draft copyrightMD schema14. It should be noted that the proposed uses of copyrightMD and PREMIS Rights are rather different. The copyrightMD schema is intended to document factual information to allow a human being to make an informed copyright assessment of a given work. The PREMIS rightsStatement is intended to allow a preservation repository to determine whether it has the right to perform a certain action in an automated fashion, with some documentation of the basis for the assertion. The first version of PREMIS limited the Rights Entity to expressing permissions held by an Agent and granted to the preservation repository.

Version 2.2 added semantic units to rightsStatement to specify a rights basis other than copyright, license or statute (e.g. institutional policy); to be able to link to further information about the rights through a documentation identifier; to associate applicable dates with a Rights statement; and to allow for term of restriction in addition to the existing term of grant. PREMIS has thus long been able to model a restriction on Rights, such as an embargo restriction on dissemination.

Version 3 contains no new additions or changes related to Rights Statements.

Version 4 introduces some major changes to Rights in order to support a broader use of the concept of Rights. In addition to the permission, there can also be the need to model an obligation; in stead of grantedRight the label *rule* is introduced, to cover both types (duty or permission). Consequently the Agent may be used to express the 'target' to whom the rule applies. In particular, the 'target' would not always be implicitly the repository. It could be a human or software Agent. As such it might be part of the repository, but doesn't have to be.

Secondly, in order to create more machine-actionability an extension is added to support coded text that describes the action, possibly in combination with restriction(s). And the option to define a trigger that activates the action, optional for a period of time (duration). These rules are defined without specific dates, as those may yet be unknown and will most likely differentiate per object. Therefore the dates are modelled in the *linkingObject* semantic unit.

Examples:

- SLAs (Service Level Agreements) or other commitments define among other things the storage media type, the transfer mode, the SIPs' maximum size, etc. So the rightsStatement based on a SLA/contract (rightsBasis) obliges the Archive to perform preservation actions.
- There is a need to specify the relationship between the format and its associated tools (characterization, validation, rendering, migration, etc.), a rightsStatement modeling a duty imposed on the repository to apply Institutional Policy (rightsBasis) to an Object.
- An Access rule (rightsStatement) based on an obligation required by a statute or copyright (rightsBasis) grants the repository the right to perform a certain action on multiple objects, but also simultaneously restricts dissemination because the prohibition rule describes an embargo duration, recorded in a duration/time-span of years without a specific start/endDate.

Of note, statutes may also represent the cultural and statutory rights of indigenous and tribal peoples to determine who may perform preservation actions and to which groups objects may be disseminated.

General Topics on the Structure and Use of the Data Dictionary

The semantic units defined in the PREMIS Data Dictionary are bound together by a few structural conventions that help organize the Data Dictionary and support its implementation. These conventions include the use of identifiers; the manner in which relationships are handled in the Data Dictionary; and the "1:1 Principle" relating metadata to Objects.

Identifiers

…

Relationships between Objects

…

Relationships between Rights

In previous iterations of PREMIS there was no way to track the history of RightsStatements (R) or relationships between them, such as changes to Rights over time that must be recorded, or that are dependent on one another or trigger conditions. A new linkingRightsIdentifier has been added which points to each applicable rightsEntity. The rightsEntity may have more than one rightsStatement.

This linking between rightsStatements can be applied in a situation where there is a dependency. For instance to reference rightStatements that can overrule a current rightsStatement. For example, under copyright the permission to provide access to the objects will be restricted by the copyright legislation. A license may lift the restrictions set by copyright. A second rightsStatement is needed. This statement describes the rights granted by the rightsholder (possibly the same Agent as is linked to the copyright-statement). In this case the license statement has priority over the copyright statement.

This may also be used to track changing rights related to trigger situations. For example, the initial rightsStatement is based on contractual agreement with a publisher; a second rightsStatement based on a contractual agreement with publisher goes into effect when trigger conditions are met. It allows the repository to assert its amended rights when the condition is met: the second rightsStatement overrules the first and the history of the rights over time is documented.

Relationships between entities of different types

The data model diagram uses arrows to show relationships between entities of different types. Objects are related to Events, Agents are related to Events, etc. The Data Dictionary expresses relationships as linking information by including in the information for entity A, a pointer to the related entity B. Every entity in the data model has a unique identifier, which can be used in a pointer. So, for example, the Object entity has arrows pointing to Rights and Events. These are implemented in the Data Dictionary by the semantic units linkingRightsStatementIdentifier and linkingEventIdentifier.

It is understood that, when a relationship between entities of different types is bi-directional, it can be stated from entity of type A to entity of type B or in the opposite direction, depending on what seems best suited to implementers. Nevertheless, some relationships are more expressive from one of the entities - for example, if one wants to express that an Object was used as documentation in an Event, they will need to record that in the Event *linkingObjectIdentifier* semantic container, as the Object *linkingEventIdentifier* semantic container has no semantic unit to express the role of the Object in the Event.

The 1:1 principle

Implementation Considerations

PREMIS conformance …

Implementation of the data model …

Storing metadata …

Supplying metadata values …

Extensibility

For several semantic units the Data Dictionary notes the potential for extensibility, to allow implementations to include additional local metadata or to provide additional structure or granularity of metadata, if required. The inclusion of such additional metadata is relatively simple for implementations using relational databases; however, a mechanism for including such metadata when using the PREMIS schemas was not available in the first release of the Data Dictionary and schemas. Version 2.0 of the Data Dictionary introduced a formal mechanism for extensibility within the schemas for a small number of semantic units which were deemed prime candidates for extension. Version ~~3.0 has~~ added the following extensible semantic units: eventDetailInformation and environmentDesignation. Version 4.0 adds another extensible semantic unit: *actionextension*. ~~Later revisions of the Data Dictionary may add to this initial set of extensible semantic units if warranted.~~

The semantic units for which extensibility is supported in the schemas are:

• significantProperties [Object entity]

• objectCharacteristics [Object entity]

• creatingApplication [within objectCharacteristics, Object entity]

• environmentDesignation [within Environment, Object entity]

• environment [within Object entity]

• signatureInformation [Object entity]

• eventDetailInformation [Event entity]

• eventOutcomeDetail [within eventOutcomeInformation, Event entity]

• rights [Rights entity]

• action [within rightsStatement, Rights entity]

• agent [Agent entity]

…

Date and time formats in PREMIS

end of P1: … The following are semantic units that may include a date or date and time:

• preservationLevelDateAssigned (under preservationLevel)

• dateCreatedByApplication (under creatingApplication)

• eventDateTime (under Event)

• startDate and endDate (under applicableDates, within rightsInformation)

• startDate and endDate (under applicableDatesOfRule, within linkingObjectIdentifer, Rights)

- startDate and endDate (under applicableDatesOfRestriction, within linkingObjectIdentifer, Rights)

- ~~copyrightStatusDeterminationDate (under copyrightInformation)~~

- ~~statuteInformationDeterminationDate (under statuteInformation)~~

- ~~startDate (under statuteApplicableDates, copyrightApplicableDates, otherRightsApplicableDates, termOfGrant, and termOfRestriction)~~

- ~~endDate (under statuteApplicableDates, copyrightApplicableDates, otherRightsApplicableDates, termOfGrant, and termOfRestriction)~~

The PREMIS Data Dictionary version 4

…

Limits to the scope of the Data Dictionary

Descriptive metadata…

Agents…

Rights: PREMIS ~~primarily~~ defines characteristics of Rights and permissions, or obligations concerned with preservation activities, ~~not~~ including those associated with access and/or distribution. The semantic units allow for extensibility to use an external Rights metadata scheme.

Technical metadata…

Media details…

Business rules: ~~The working group made no attempt to describe the business rules of a repository, although certainly this metadata is essential for preservation within the repository~~ Business rules codify the application of preservation strategies and document repository policies, services, charges, and roles. Retention periods, disposition, risk assessment, permanence ratings, schedules for media refreshment, and so on are pertinent to objects but are not actual properties of Objects. A single exception was made for the level of preservation treatment to be accorded an object (preservationLevel) because this was felt to be critical information for any preservation repository. ~~A more thorough treatment of business rules could be added to the data model by defining a Rules entity similar to Rights, although this is not included in the current revision.~~ Version 4 introduces a new approach for the treatment of business rules, which can now be implemented by defining a rule (type: obligation) in a rightsStatement.

Object Entity

The Object entity aggregates information about a digital object held by a preservation repository and describes those characteristics relevant to preservation management.

The only mandatory semantic units that apply to all categories of Object (Intellectual Entity, Representation, File, and Bitstream) are objectIdentifier and objectCategory.

Entity types …

Entity properties

- Can be associated with one or more Rights statements.
- Can participate in one or more Events.

Links between entities may be recorded from either direction and need not be bi-directional.

Entity semantic units

…

1.15 linkingRightsStatementIdentifier (O, R)

1.15.1 linkingRights~~Statement~~IdentifierType (M, NR)

1.15.2 linkingRights~~Statement~~IdentifierValue (M, NR)

Event Entity

The Event entity aggregates information about an action that involves one or more Object entities. Metadata about an Event would normally be recorded and stored separately from the digital object.

Whether or not a preservation repository records an Event depends upon the importance of the event. Actions that modify objects should always be recorded. Other actions such as copying an object for backup purposes may be recorded in system logs or an audit trail but not necessarily in an Event entity.

Mandatory semantic units are: eventIdentifier, eventType, and eventDateTime.

Entity properties

- Must be related to one or more Objects.
- Can be related to one or more Agents.
- Can be related to Rights entities as review events for rights status.

Links between entities may be recorded from either direction and need not be bi-directional.

The Event entity may be pointed to from a linking relation between Object and Rights. In this case the Rights of that particular Object (or Objects) are related to the event. For example when, as an outcome of a Review, the rightsStatement has been changed, or newly created.

Entity semantic units

2.1 eventIdentifier (M, NR)
  2.1.1 eventIdentifierType (M, NR)
  2.1.2 eventIdentifierValue (M, NR)
2.2 eventType (M, NR)
2.3 eventDateTime (M, NR)
2.4 eventDetailInformation (O, R)
  2.4.1 eventDetail (O, NR)

2.4.2 eventDetailExtension (O, R)
2.5 eventOutcomeInformation (O, R)
2.5.1 eventOutcome (O, NR)
2.5.2 eventOutcomeDetail (O, R)
2.5.2.1 eventOutcomeDetailNote (O, NR)
2.5.2.2 eventOutcomeDetailExtension (O, R)
2.6 linkingAgentIdentifier (O, R)
2.6.1 linkingAgentIdentifierType (M, NR)
2.6.2 linkingAgentIdentifierValue (M, NR)
2.6.3 linkingAgentRole (O, R)
2.7 linkingObjectIdentifier (O, R)
2.7.1 linkingObjectIdentifierType (M, NR)
2.7.2 linkingObjectIdentifierValue (M, NR)
2.7.3 linkingObjectRole (O, R)

Agent Entity

The Agent entity aggregates information about attributes or characteristics of Agents (persons, organizations, or software) associated with Rights management and preservation events in the life of a data object. Agent information serves to identify an Agent unambiguously from all other Agent entities.

An agent is a person, organization, hardware, or software, or may also be a group (an organizational unit or a class of staff with some shared characteristic, such as all staff in a preservation unit or all staff with a specific security clearance). A note on the word "repository": the term is sometimes used to refer to an organization, and sometimes refers to repository software or a system.

A piece of software or hardware that is captured as an Agent can also be preserved in a repository and described as an environment Object, for example, as source code or an ISO disk image. In this case, implementers may choose to relate the Agent to the environment Object using the linkingEnvironmentIdentifier semantic unit. This relationship can support the ability of a repository to record in considerable detail the interactions between software or hardware Agents and the preserved digital objects with which they interact, and even to reproduce these interactions if desired.

The only mandatory semantic unit is agentIdentifier.

Entity properties

- May hold or grant one or more Rights (resp. be committed to or impose an obligation).
- May carry out, authorize, or compel one or more Events.
- May create or act upon one or more Objects through an Event or with respect to a Rights statement.

Entity semantic units

3.1 agentIdentifier (M, R)
- 3.1.1 agentIdentifierType (M, NR)
- 3.1.2 agentIdentifierValue (M, NR)

3.2 agentName (O, R)
3.3 agentType (O, NR)
3.4 agentVersion (O, NR)
3.5 agentNote (O, R)
3.6 agentExtension (O, R)
3.7 linkingEventIdentifier (O, R)
- 3.7.1 linkingEventIdentifierType (M, NR)
- 3.7.2 linkingEventIdentifierValue (M, NR)

3.8 linkingRights~~Statement~~Identifier (O, R)
- 3.8.1 linkingRights~~Statement~~IdentifierType (M, NR)
- 3.8.2 linkingRights~~Statement~~IdentifierValue (M, NR)

3.9 linkingEnvironmentIdentifier (O, R)

3.9.1 linkingEnvironmentIdentifierType (M, NR)
3.9.2 linkingEnvironmentIdentifierValue (M, NR)
3.9.3 linkingEnvironmentRole (O, R)

Rights Entity

For the purpose of the PREMIS Data Dictionary, statements of legal rights, ~~and~~ permissions and obligations are taken to be constructs that can be described as the Rights entity. Rights are entitlements granted to Agents by copyright or other intellectual property law. Permissions are powers or privileges granted by agreement between a rightsholder (the Agent) and another party or parties (being the repository). In fact it can even be the repository that grants rights to another party or parties (Agents).

Obligations are also based on such agreements, but with a binding character. And in that case the Agent might be an instrument (software, hardware) or a contract partner (organisation, person etc).

In the situation that the copyright belongs to a third party (the rights holder), permission can be granted through an agreement (a license) to perform actions for which copyright is required. The Repository might want to record the duration of the license as well as when the copyright as such expires, thus ending the need for licensing. Especially when permission is generic, rather than relating to a specific request or a one-time action. The Rights Entity of the (generic) license can be linked to the Rights Entity of the copyright, to make their relation explicit.

A repository might wish to record a variety of Rights information including abstract Rights statements and statements of permissions that apply to external Agents and to objects not held within the repository. The minimum core Rights information that a preservation repository must know, however, is what Rights, ~~or~~ being permissions or obligations, a repository has, to carry out actions related to objects within the repository. These may generally be granted by copyright law, by statute, or by a (license) agreement with the rightsholder or another third party. In some situations the basis for the rights is for other reasons, for instance institutional policy.

If the repository records Rights information, either *rightsStatement* or *rightsExtension* must be present.

Entity properties

- Rights entities may be related to one or more Objects, with one or more Events.
- Rights entities may be related to one or more Agents.
- Rights entities may be related to one or more Rights.

Links between entities may be recorded from either direction and need not be bi-directional.

Entity semantic units

4.1 rightsIdentifier (M, R)
- 4.1.1 rightsIdentifierType (M, NR)
- 4.1.2 rightsIdentifierValue (M, NR)

4.2 rightsStatement (O,R)
- 4.2.1 rightsBasis (M, NR)
- 4.2.2 rightsInformation (O, NR)
  - 4.2.2.1 jurisdiction (O, NR)
  - 4.2.2.2 conditions (O, NR)
  - 4.2.2.3 citation (O, NR)
  - 4.2.2.4 note (O, R)
  - 4.2.2.5 documentationIdentifier (O, R)
    - 4.2.2.5.1 documentationIdentifierType (M, NR)
    - 4.2.2.5.2 documentationIdentifierValue (M, NR)
    - 4.2.2.5.3 documentationRole (O, NR)
  - 4.2.2.6 applicableDates (O, NR)
    - 4.2.2.6.1 startDate (O, NR)
    - 4.2.2.6.2 endDate (O, NR)
  - 4.2.2.7 otherRightsBasis (M, NR)
- 4.2.3 rule (O, R)
  - 4.2.3.1 ruletype (M, NR)
  - 4.2.3.2 action (M, NR)
    - 4.2.3.2.1 actiontype (M, NR)
    - 4.2.3.2.2 actionextension (O, NR)
  - 4.2.3.3 restriction (O, R)
    - 4.2.3.3.1 restrictionType (M, NR)
    - 4.2.3.3.2 termOfRestriction (O, NR)
    - 4.2.3.3.2.1 trigger (O, NR)
    - 4.2.3.3.2.2 duration (O, NR)
  - 4.2.3.4 termOfRule (O, NR)
    - 4.2.3.4.1 trigger (O, NR)
    - 4.2.3.4.2 duration (O, NR)
  - 4.2.3.5 ruleNote (O, R)
- 4.2.4 linkingObjectIdentifier (O, R)
  - 4.2.4.1 linkingObjectIdentifierType (M, NR)
  - 4.2.4.2 linkingObjectIdentifierValue (M, NR)
  - 4.2.4.3 linkingObjectRole (O, R)
  - 4.2.4.4 applicableDatesOfRule (O, R)
    - 4.2.4.4.1 startDate (O, NR)
    - 4.2.4.4.2 endDate (O, NR)
  - 4.2.4.5 applicableDatesOfRestriction (O, R)

4.2.4.5.1 startDate (O, NR)

4.2.4.5.2 endDate (O, NR)

~~4.2.4.6 relatedAgentIdentifier (O, R)~~

~~4.2.4.6.1 relatedAgentIdentifierType (M, NR)~~

~~4.2.4.6.2 relatedAgentIdentifierValue (M,NR)~~

~~4.2.4.6.3 relatedAgentRole (O, R)~~

4.2.4.6 relatedEventIdentifier (O, R)

4.2.4.7.1 relatedEventIdentifierType (M, NR)

4.2.4.7.2 relatedEventIdentifierValue (M, NR)

4.2.4.7.3 relatedEventType (O, R)

4.3 rightsExtension (O, R)

4.4 linkingAgentIdentifier (O, R)

4.4.1 linkingAgentIdentifierType (M, NR)

4.4.2 linkingAgentIdentifierValue (M, NR)

4.4.3 linkingAgent Role (O, R)

4.5 linkingRightsIdentifier (O, R)

4.5.1 linkingRightsIdentifierType (M, NR)

4.5.2 linkingRightsIdentifierValue (M, NR)

4.5.3 linkingRightsRole (O, R)

Semantic unit	4.1 rights~~Statement~~Identifier
Semantic components	4.1.1 rightsIdentifierType (M, NR) ~~rightsStatementIdentifierType~~ 4.1.2 rightsIdentifierValue (M, NR) ~~rightsStatementIdentifierValue~~
Definition	The designation used to uniquely identify the Rights within a preservation repository system
Rationale	Each statement of rights associated with the preservation repository must have a unique identifier to allow it to be related to events and agents.
Data constraint	Container
Repeatability	Not repeatable
Obligation	Mandatory
Usage notes	Identifiers must be unique within the repository.

Semantic unit	4.1.1 rights~~Statement~~IdentifierType
Semantic components	None
Definition	A designation of the domain in which the Rights identifier is unique.
Data constraint	Value should be taken from a controlled vocabulary. A controlled vocabulary is available at: http://id.loc.gov/vocabulary/identifiers.html.
Example	LCNAF SAN DLC URI local
Repeatability	Not Repeatable
Obligation	Mandatory

Semantic unit	4.1.2 rights~~Statement~~IdentifierValue

Semantic components	None
Definition	The value of the *rightsIdentifier* ~~*rightsStatementIdentifier*~~
Data constraint	Value may be taken from a controlled vocabulary.
Example	
Repeatability	Not Repeatable
Obligation	Mandatory
Usage notes	May be a unique key or a controlled textual form of name

Semantic unit	**4.2 rightsStatement**
Semantic components	~~4.1.1 rightsStatementIdentifier~~ ~~4.1.2 rightsBasis~~ ~~4.1.3 copyrightInformation~~ ~~4.1.4 licenseInformation~~ ~~4.1.5 statuteInformation~~ ~~4.1.6 otherRightsInformation~~ ~~4.1.7 rightsGranted~~ ~~4.1.8 linkingObjectIdentifier~~ ~~4.1.9 linkingAgentIdentifier~~ 4.2.1 rightsBasis (M, NR) 4.2.2 rightsInformation (O, NR) 4.2.3 rule (O, R) 4.2.4 linkingObjectIdentifier (O, R)
Definition	Documentation of the repository's Rights (duty or obligation) or indeed restrictions to perform one or more actions.
Data constraint	Container
Repeatability	Repeatable
Obligation	Optional
Usage notes	This semantic unit is optional because in some cases Rights may be unknown. Institutions are encouraged to record Rights information when possible. Either *rightsStatement* or *rightsExtension* must be present if the Rights entity is included.

	The *rightsStatement* should be repeated when the act(s) described has (have) more than one basis, or when different acts have different bases.

Semantic unit	4.2.1 rightsBasis
Semantic components	None
Definition	Designation of the basis for the right, (permission or obligation) identified by the *rights~~Statement~~Identifier*.
Data constraint	Values should be taken from a controlled vocabulary. A controlled vocabulary is available at: http://id.loc.gov/vocabulary/preservation/rightsBasis.html.
Repeatability	Not repeatable
Obligation	Mandatory
Examples	copyright license statute other
Usage Notes	~~When rightsBasis is "copyright", copyrightInformation should be provided.~~ ~~When rightsBasis is "license", licenseInformation should be provided.~~ ~~When rightsBasis is "statute", statuteInformation should be provided.~~ When rightsBasis is "other", otherRightsBasis (in ~~otherRightsInformation~~ rightsInformation container) should be provided. If the Rights for the item are public domain, use "copyright". If they are Fair Use, use "statute". If more than one basis applies, the entire Rights entity should be repeated.

Semantic unit	~~**4.1.3 copyrightInformation**~~ ~~**4.1.4 licenseInformation**~~ ~~**4.1.5 statuteInformation**~~ ~~**4.1.6 otherRightsInformation**~~ **4.2.2 rightsInformation**
Semantic components	~~4.1.3.1 copyrightStatus~~ ~~4.1.3.2 copyrightJurisdiction~~ ~~4.1.3.3 copyrightStatusDeterminationDate~~ ~~4.1.3.4 copyrightNote~~ ~~4.1.3.5 copyrightDocumentationIdentifier~~ ~~4.1.3.6 copyrightApplicableDates~~ ~~4.1.4.1 licenseDocumentationIdentifier~~ ~~4.1.4.2 licenseTerms~~ ~~4.1.4.3 licenseNote~~ ~~4.1.4.4 licenseApplicableDates~~ ~~4.1.5.1 statuteJurisdiction~~ ~~4.1.5.2 statuteCitation~~ ~~4.1.5.3 statuteInformationDeterminationDate~~ ~~4.1.5.4 statuteNote~~ ~~4.1.5.5 statuteDocumentationIdentifier~~ ~~4.1.5.6 statuteApplicableDates~~ ~~4.1.6.1 otherRightsDocumentationIdentifier~~ ~~4.1.6.2 otherRightsBasis~~ ~~4.1.6.3 otherRightsApplicableDates~~ ~~4.1.6.4 otherRightsNote~~ 4.2.2.1 jurisdiction 4.2.2.2 conditions 4.2.2.3 citation 4.2.2.4 note 4.2.2.5 documentationIdentifier 4.2.2.6 applicableDates 4.2.2.7 otherRightsBasis
Definition	~~Information about the copyright status of the object(s).~~

	~~Information about a license or other agreement granting permissions related to an object.~~ ~~Information about the statute allowing use of the object.~~ ~~Information about the Rights (other than copyright, license, or statute) that apply to the object(s).~~ Information about the Rights that apply to the object(s): . about the copyright status . about a license or other agreement granting permissions or imposing obligations related to an object . about a statute allowing use, or imposing obligations to the object . about another rightbasis than the above
Data constraint	Container
Repeatability	Not repeatable
Obligation	Optional
Usage notes	~~When rightsBasis is "copyright", copyrightInformation should be provided.~~ When rightsBasis is "copyright", repositories may need to extend this with more detailed information. See the California Digital Library's copyrightMD schema (http://www.cdlib.org/groups/rmg//) for an example of a more detailed scheme.

Semantic unit	**4.2.2.1 jurisdiction**
Semantic components	None
Definition	The country whose copyright-laws apply or the country or other political body enacting the statute. This may also document statutes covered by indigenous and tribal sovereignty.
Rationale	Copyright law/Statutes can vary from country to country. The connection between the object and the Rights granted is based on jurisdiction.
Data constraint	Values should be taken from ISO 3166.
Example	us de
Repeatability	Not repeatable

Obligation	Optional (Mandatory in case rightsBasis = copyright or statute)

Semantic unit	**4.2.2.2 conditions**
Semantic components	None
Definition	Text describing the license or agreement by which permission was granted.
Data constraint	None
Repeatability	Not repeatable
Obligation	Optional
Usage notes	This could contain the actual text of the license or agreement, or a paraphrase or summary of it. Most relevant in combination with rightsBasis is license or other.

Semantic unit	**4.2.2.3 citation**
Semantic components	None
Definition	An identifying designation for the statute.
Data constraint	None
Example	Legal Deposit (Jersey) Law 200- National Library of New Zealand (Te Puna Mātauranga o Aotearoa) Act 2003 no 19 part 4 s 34
Repeatability	Not repeatable
Obligation	Optional (Mandatory when rightsBasis = statute)
Usage notes	Use standard citation form when applicable.

Semantic unit	4.2.2.4 note
Semantic components	None
Definition	Additional information about Rights of the object.
Data constraint	None
Examples	Copyright expiration expected in 2010 unless renewed. Copyright statement is embedded in the file header. License is embedded in XMP block in file header. Applicability to web-published content sent for review by general counsel 9/19/2008. 80-year rule
Repeatability	Repeatable
Obligation	Optional
Usage notes	Information about the terms of the license should go in conditions ~~*licenseTerms*~~. *Note* is intended for other types of information related to the license/agreement, such as contact persons, action dates, or interpretations. The note may also indicate the location of the license, for example, if it is available online or embedded in the object itself.

Semantic unit	4.2.2.5 documentationIdentifier
Semantic components	4.2.2.7.1 documentationIdentifierType 4.2.2.7.2 documentationIdentifierValue 4.2.2.7.3 documentationRole
Definition	A designation used to identify documentation supporting the specified Rights within the repository system granted uniquely . according to copyright . granted by license . granted by statute . or when the basis is something other than these three
Data constraint	Container
Repeatability	Repeatable
Obligation	Optional
Usage notes	This semantic unit is intended to refer to a document . recording the granting of permission, the expression of requirements or restrictions, or other information supporting the specified *rightsBasis*. . in case of license/agreement: this may be a formal signed contract with a customer. If the granting agreement is verbal, this could point to a memo by the repository documenting the verbal agreement. The identifier is optional because the agreement may not be stored in a repository with an identifier. For example, in the case of a verbal agreement the entire agreement may be included or described in the *conditions*. If repeated, use *documentationRole* to distinguish the role of the given documentation.

Semantic unit	**4.2.2.5.1 documentationIdentifierType**
Semantic components	None
Definition	A designation of the domain within which the documentation identifier is unique.
Data constraint	Value should be taken from a controlled vocabulary. A controlled vocabulary is available at: http://id.loc.gov/vocabulary/identifiers.
Repeatability	Not repeatable
Obligation	Mandatory

Semantic unit	**4.2.2.5.2 documentationIdentifierValue**
Semantic components	None
Definition	The value of the *documentationIdentifier*.
Data constraint	None
Repeatability	Not repeatable
Obligation	Mandatory

Semantic unit	**4.2.2.5.3 documentationRole**
Semantic components	None
Definition	A value indicating the purpose or expected use of the documentation being identified.
Rationale	This information distinguishes the purpose of the supporting documentation especially when there are multiple documentation identifiers.
Examples	accession record copyright statement donor agreement law application decree case law institutional policy

	deed of gift
Data constraint	Value should be taken from a controlled vocabulary.
Repeatability	Not repeatable
Obligation	Optional
Usage notes	For a particular law one might want to link to various sources of documentation, e.g. the law publication itself (role: law), the application decree that enforces i (rol: application decree) or some additional text refining the law by showing a real world verdict (role: case law).

Semantic unit	**4.2.2.6 applicableDates**
Semantic components	4.2.2.8.1 startDate 4.2.2.8.2 enddate
Definition	The date range during which the particular *rightsBasis* applies or is applied to the content. This is distinct from ~~termOfGrant~~ *the applicableDatesOfRule*, which applies to a particular act expressed in ~~rightsGranted~~ *rule and to a specific (set of) objects*. and may differ from the period of time the license, statute, or other basis applies to the content.
Rationale	Specific dates may apply to the particular Rights basis ~~granted~~.
Data constraint	Container
Repeatability	Not repeatable
Obligation	Optional
Usage notes	The repository may wish to retain the history of Rights and restrictions associated with the content over time. Associating active dates with particular Rights bases allows applications to identify which of several *rightsStatements* are in force at a given time. Most relevant in combination with rightsBasis is statute, license or other. In case of copyright the applicable dates usually depend on context of the object; the applicable dates in general would be the start-enddate of the copyright law.

Semantic unit	**4.2.2.6.1 startDate**

Semantic components	None
Definition	The date the Rights basis commences.
Data constraint	To aid machine processing, value should use a structured form. To facilitate exchange of PREMIS-conformant metadata, use of standard conventions, for instance as used in the date elements in the PREMIS schema, is recommended.
Examples	2006-01-02 20050723
Repeatability	Not repeatable
Obligation	Optional

Semantic unit	**4.2.2.6.2 endDate**
Semantic components	None
Definition	The date the ~~granted copyright~~ Rights basis expires.
Data constraint	To aid machine processing, value should use a structured form. To facilitate exchange of PREMIS-conformant metadata, use of standard conventions, for instance as used in the date elements in the PREMIS schema, is recommended.
Examples	2010-01-02 20120723
Repeatability	Not repeatable
Obligation	Optional
Usage notes	Use “OPEN” for an open ended term of restriction. Omit *endDate* if the ending date is unknown or the permission statement applies to many objects with different end dates.

Semantic unit	**4.2.2.7 otherRightsBasis**
Semantic components	None

Definition	Designation of the basis for the other right, ~~or~~ permission or obligation described in the *rightsStatement*.
Data constraint	Values should be taken from a controlled vocabulary.
Example	Harvard policy
Repeatability	Not repeatable
Obligation	Mandatory
Usage notes	Use this semantic unit for specific Rights bases other than *copyright*, *license* or, *statute*. When this semantic unit is used, 4.2.1 *rightsBasis* should be set to "other". The Rights basis may be specific to the repository, but it is recommended to use a value from a local or globally controlled vocabulary for machine actionability. If more than one basis applies, the entire Rights entity should be repeated.

Semantic unit	**4.2.3 rule**
Semantic components	4.2.3.1 ruleType 4.2.3.2 action 4.2.3.3 restriction 4.2.3.4 termOfRule 4.2.3.5 ruleNote
Definition	Statement about the actions a repository has permission or the duty to undertake or is prohibited from undertaking with respect to an Object.
Data constraint	Container
Repeatability	Repeatable
Obligation	Optional

Semantic unit	**4.2.3.1 ruleType**
Semantic components	None

Definition	A categorization of the nature of the Rule.
Rationale	Categorizing Rules will aid the preservation repository in machine processing of rule information, particularly as a sorting mechanism for reporting.
Data constraint	Value should be taken from a controlled vocabulary.
Examples	duty ~~obligation??~~ permission
Repeatability	Not repeatable
Obligation	Mandatory
Usage notes	The Rule defines the class of actions that a repository is allowed to perform. A duty is an obligation–which may be a restriction–and both a duty or permission may be granted by statute or license as documented in rightsBasis or institutional policy or contract as documented in otherRightsBasis.

Semantic unit	**4.2.3.2 action**
Semantic components	**4.2.3.2.1 actionType** **4.2.3.2.2 actionExtension**
Definition	The action the preservation repository is allowed (prohibited/obliged) to take.
Data constraint	Container
Repeatability	Not repeatable
Obligation	Mandatory

Semantic unit	**4.2.3.2.1 actionType**
Semantic components	None
Definition	The action the preservation repository is allowed (prohibited/obliged) to take.

Data constraint	Value should be taken from a controlled vocabulary. A controlled vocabulary is available at: http://id.loc.gov/vocabulary/preservation/actionsGranted.html
Examples	Replicate Modify Use Disseminate
Repeatability	Not repeatable
Obligation	Mandatory
Usage notes	It is up to the preservation repository to decide how granular the controlled vocabulary should be. It may be useful to employ the same controlled values that the repository uses for *eventType*.

Semantic unit	**4.2.3.2.2 actionExtension**
Semantic components	None
Definition	The action ('act') is optionally extended to add machine readable instructions.
Data constraint	
Examples	
Repeatability	Not repeatable
Obligation	Optional
Usage notes	

Semantic unit	**4.2.3.3 restriction**
Semantic components	**4.2.3.3.1 restrictionType** **4.2.3.3.2 termOfRestriction**
Definition	A condition or limitation on the act.
Data constraint	Container

Examples	No more than three Allowed only after one year of archival retention has elapsed Rightsholder must be notified after completion of act
Repeatability	Repeatable
Obligation	Optional

Semantic unit	**4.2.3.3.1 restrictionType**
Semantic components	None
Definition	A categorization of the nature of the Restriction.
Rationale	Categorizing restrictions will aid the preservation repository in machine processing of restriction information, particularly as a sorting mechanism for reporting.
Data constraint	Value should be taken from a controlled vocabulary.
Examples	embargo security classification
Repeatability	Not repeatable
Obligation	Mandatory
Usage notes	The Rule defines the class of actions that a repository is allowed to perform. A duty is an obligation–which may be a restriction–and both a duty or permission may be granted by statute or license as documented in rightsBasis or institutional policy or contract as documented in otherRightsBasis.

Semantic unit	**4.2.3.3.2 termOfRestriction**
Semantic components	4.2.3.3.2.1 restriction Trigger ~~startDate~~ 4.2.3.3.2.2 restrictionDuration ~~endDate~~
Definition	The time period for the restriction granted.
Rationale	~~The current definition of termOfGrant is "time period for the permissions granted."~~ This allows for expressing information about ~~the rights granted, but some repositories may need to express~~

	time-bounded restrictions like embargoes. If this is applicable, use~~startDate~~ trigger to identify the type of start date of enforcement and ~~endDate~~ duration for the end of the ~~embargo~~ time period should be recorded.
Data constraint	Container
Repeatability	Not repeatable
Obligation	Optional

Semantic unit	**4.2.3.3.2.1 restrictionTrigger**
Semantic components	None
Definition	~~The time period for the permissions granted.~~ The type of start date or startdate-time that launches the enforcement period.
Data constraint	Use a local controlled vocabulary to identify the type of trigger. ~~To aid machine processing, value should use a structured form. To facilitate exchange of PREMIS-conformant metadata, use of standard conventions, for instance as used in the date elements in the PREMIS schema, is recommended.~~
Examples	Date of Classification Donor Agreement Signed Licensing Date Date of birth Date of death Date of publication
Repeatability	Not repeatable
Obligation	Optional

Semantic unit	4.2.3.3.2.2 **restrictionDuration**
Semantic components	None
Definition	The time duration for enforcement. ~~rule enforcement~~

Data constraint	To aid machine processing, value should use a structured form. To facilitate exchange of PREMIS-conformant metadata, use of standard conventions, for instance as used in the date elements in the PREMIS schema, is recommended.
Examples	P5Y P6M OPEN ~~2010-01-02~~ ~~20120723~~
Repeatability	Not repeatable
Obligation	Optional, Mandatory if there is a trigger
Usage notes	Use “OPEN” for an open ended term of grant.

Semantic unit	**4.2.3.4 termOfRule**
Semantic components	4.2.3.4.1 ruleTrigger ~~startDate~~ 4.2.3.4.2 ruleDuration ~~endDate~~
Definition	The type of trigger that launches the period of rule enforcement and the duration of the period.
Rationale	~~The permission to preserve or restrictions may be time bounded~~ A rule–permission or obligation–may have a specific trigger and duration, such as a term of copyright after publication.
Data constraint	Container
Repeatability	Not repeatable
Obligation	Optional

Semantic unit	**4.2.3.4.1 ruleTrigger ~~startDate~~**
Semantic components	None
Definition	~~The time period for the permissions granted.~~ The type of start date or startdate-time that launches the enforcement period.
Data constraint	Use a controlled vocabulary to identify the category of trigger.

	~~To aid machine processing, value should use a structured form. To facilitate exchange of PREMIS-conformant metadata, use of standard conventions, for instance as used in the date elements in the PREMIS schema, is recommended.~~
Examples	Date of Classification Donor Agreement Signed Licensing Date Date of birth Date of death Date of publication
Repeatability	Not repeatable
Obligation	Optional

Semantic unit	**4.2.3.4.2 ruleDuration ~~endDate~~**
Semantic components	None
Definition	The time duration for enforcement.
Data constraint	To aid machine processing, value should use a structured form. To facilitate exchange of PREMIS-conformant metadata, use of standard conventions, for instance as used in the date elements in the PREMIS schema, is recommended.
Examples	P5Y P6M OPEN ~~2010-01-02~~ ~~20120723~~
Repeatability	Not repeatable
Obligation	Optional, Mandatory if there is a Trigger
Usage notes	Use “OPEN” for an open ended term of grant.

Semantic unit	**4.2.3.5 ruleNote**
Semantic components	None

Definition	Additional information about the rule.
Rationale	A textual description of the rule may be needed for additional explanation.
Data constraint	None
Repeatability	Repeatable
Obligation	Optional
Usage notes	This semantic unit may include a statement about risk assessment, for example, when a repository is not certain about what aspects of a rule should be enforced.

Semantic unit	**4.2.4 linkingObjectIdentifier**
Semantic components	4.2.4.1 linkingObjectIdentifierType 4.2.4.2 linkingObjectIdentifierValue 4.2.4.3 linkingObjectRole
Definition	The identifier of an object associated with the Rights statement.
Rationale	Rights statements should be associated with the objects to which they pertain, either by linking from the Rights statement to the object(s) or by linking from the object(s) to the Rights statement. This semantic unit provides the mechanism for the link from the Rights statement to an object. Each rule can be connected to Objects via the linking element with the Object-specific dates in place. If this is used, then the linkingRightsStatementIdentifier of the ObjectEntity should also be used because these linking elements should be each other's mirror image.
Data constraint	Container
Repeatability	Repeatable
Obligation	Optional

Usage notes	Linking semantic units are mandatory in the sense that a repository needs to know the information, but are defined as optional because PREMIS does not specify in which direction the linkage should be. In particular, *linkingObjectIdentifier* is optional because in some cases it will be more practical to link from the object(s) to the Rights statement; for example, a repository may have a single Rights statement covering thousands of public domain objects.

Semantic unit	**4.2.4.1 linkingObjectIdentifierType**
Semantic components	None
Definition	A designation of the domain in which the linking object identifier is unique.
Data constraint	Must be an existing *objectIdentifierType* value.
Examples	[see examples for *objectIdentifierType*]
Repeatability	Not repeatable
Obligation	Mandatory

Semantic unit	**4.2.4.2 linkingObjectIdentifierValue**
Semantic components	None
Definition	The value of the *linkingObjectIdentifier*.
Data constraint	Must be an existing *objectIdentifierValue* value.
Examples	[see examples for *objectIdentifierValue*]
Repeatability	Not repeatable
Obligation	Mandatory

Semantic unit	**4.2.4.3 linkingObjectRole**

Semantic components	None
Definition	The role of the object described by the rightsStatement
Rationale	Distinguishes the role of the object. If this is not a direct or explicit relationship, it is necessary to analyze the relationship in the object metadata.
Data constraint	None
Repeatability	Repeatable
Obligation	Optional
Usage notes	This value need not be supplied in the ordinary case where the role of the linked-to object is to be governed by the Rights statement. If the object has a different relationship to the Rights statement, however, it should be noted here.

Semantic unit	**4.2.4.4 applicableDatesOfRule**
Semantic components	4.2.4.4.1 startDate 4.2.4.4..2 endDate
Definition	The date range during which permissions assigned by statute apply or are applied to the Object described by the rightsStatement. This is distinct from termOfRule, which applies to a particular act expressed in rightsGranted and may differ from the period of time the license, statute or other basis applies to the content.
Rationale	Specific dates may apply to the particular rights granted.
Data constraint	Container
Repeatability	Repeatable ~~Not repeatable~~
Obligation	Optional
Usage notes	The repository may wish to retain the history of rights and restrictions associated with the content over time. Associating active dates with particular rights bases allows applications to identify which of several rightsStatements are in force at a given time.

Semantic unit	4.2.4.4.1 startDate
Semantic components	None
Definition	The beginning date for which the permission applies or is applied to the Object described by the rightsStatement.
Data constraint	To aid machine processing, value should use a structured form. To facilitate exchange of PREMIS-conformant metadata, use of standard conventions, for instance as used in the date elements in the PREMIS schema, is recommended.
Examples	2006-01-02 20050723
Repeatability	Not repeatable
Obligation	Optional

Semantic unit	4.2.4.4.2 endDate
Semantic components	None
Definition	The ending date for which the permission applies or is applied to the Object described by the rightsStatement.
Data constraint	To aid machine processing, value should use a structured form. To facilitate exchange of PREMIS-conformant metadata, use of standard conventions, for instance as used in the date elements in the PREMIS schema, is recommended.
Examples	2010-01-02 20120723
Repeatability	Not repeatable
Obligation	Optional
Usage notes	Use “OPEN” for an open ended term of restriction. Omit *endDate* if the ending date is unknown or the permission statement applies to many objects with different end dates.

Semantic unit	4.2.4.5 applicableDatesOfRestriction

Semantic components	4.2.4.5.1 startDate 4.2.4.5.2 endDate
Definition	The date range during which the restriction identified in a statute applies or is applied to the content. This is distinct from termOfRule, which applies to a particular act expressed in rightsGranted and may differ from the period of time the license, statute or other basis that applies to the content.
Rationale	Specific dates may apply to the particular rights granted.
Data constraint	Container
Repeatability	Not repeatable
Obligation	Optional
Usage notes	The repository may wish to retain the history of rights and restrictions associated with the content over time. Associating active dates with particular rights bases allows applications to identify which of several rightsStatements are in force at a given time.

Semantic unit	**4.2.4.5.1 startDate**
Semantic components	None
Definition	The beginning date for which the restriction applies or is applied to the Object described by the rightsStatement.
Data constraint	To aid machine processing, value should use a structured form. To facilitate exchange of PREMIS-conformant metadata, use of standard conventions, for instance as used in the date elements in the PREMIS schema, is recommended.
Examples	2006-01-02 20050723
Repeatability	Not repeatable
Obligation	Optional

Semantic unit	**4.2.4.5.2 endDate**
Semantic components	None

Definition	The ending date for which the restriction applies or is applied to the Object described by the rightsStatement.
Data constraint	To aid machine processing, value should use a structured form. To facilitate exchange of PREMIS-conformant metadata, use of standard conventions, for instance as used in the date elements in the PREMIS schema, is recommended.
Examples	2010-01-02 20120723
Repeatability	Not repeatable
Obligation	Optional
Usage notes	Use "OPEN" for an open ended term of restriction. Omit *endDate* if the ending date is unknown or the permission statement applies to many objects with different end dates.

Semantic unit	**4.2.4.6 relatedEventIdentifier**
Semantic components	4.2.4.6.1 relatedEventIdentifierType 4.2.4.6.2 relatedEventIdentifierValue 4.2.4.6.3 relatedEventType
Definition	A review Event takes place, which has an eventOutcome and an optional relatedAgentIdentifier and is linked to the object using linkingObjectIdentifier. IF the review event results in a partial update to the rights, a new rightsStatement is created, and is linked to the review Event. The outcome of the review could be that nothing has changed. If there is no change, an Event must still be created and linked that reconfirms the status. A rightsStatement may also be updated INSTEAD of creating a new rightsStatement in cases where the outcome of the review event is a wholesale change, such as the expiration of a license or copyright. Review Events with partial outcomes must have a new rightsStatement. The *eventIdentifier* of a review Event may be associated with an Agent using a relatedAgent Identifier with the relatedAgentRole of "Authorizer."

	If there are review Events associated with one or more Agents, one must be able to link the review Event to each of the agents, and the outcomes of multiple reviews which may relate to any number of agents over time.		
Data constraint	Container		
Object category	**Representation**		
Applicability	Applicable		
Repeatability	Repeatable		
Obligation	Optional		
Usage notes	Use to link to Events that are not associated with relationships between Objects, such as format validation, virus checking etc. If the Event involves the creation of another Object, use *relatedEventIdentifer* to link the newly-created Object to the Event, and *relatedAgentIdentifier* to link that Event to the Agent. Linking semantic units are mandatory in the sense that a repository needs to know the information, but are defined as optional because PREMIS does not specify in which direction the linkage should be.		

Semantic unit	**4.2.4.6.1 relatedEventIdentifierType**		
Semantic components	None		
Definition	The *relatedEventIdentifierType* value of the related event.		
Data constraint	Must be an existing *eventIdentifierType* value.		
Object category	**Intellectual Entity / Representation**	**File**	**Bitstream**
Applicability	Applicable	Applicable	Applicable
Examples	[see examples for *eventIdentifierType*]	[see examples for *eventIdentifierType*]	[see examples for *eventIdentifierType*]
Repeatability	Not repeatable	Not repeatable	Not repeatable
Obligation	Mandatory	Mandatory	Mandatory

Semantic unit	**4.2.4.6.2 relatedEventIdentifierValue**

Semantic components	None		
Definition	The *relatedEventIdentifierValue* value of the related event.		
Data constraint	Must be an existing *eventIdentifierValue* value.		
Object category	**Intellectual Entity / Representation**	**File**	**Bitstream**
Applicability	Applicable	Applicable	Applicable
Examples	[see examples for *eventIdentifierValue*]	[see examples for *eventIdentifierValue*]	[see examples for *eventIdentifierValue*]
Repeatability	Not repeatable	Not repeatable	Not repeatable
Obligation	Mandatory	Mandatory	Mandatory

Semantic unit	**4.2.4.6.3 relatedEventType**
Semantic components	None
Definition	The type of the related Event in relation to the Rights statement.
Data constraint	Values should be taken from a controlled vocabulary.
Examples	review
Repeatability	Repeatable
Obligation	Optional

Semantic unit	**4.3 rightsExtension**
Semantic components	Defined externally
Definition	A container to include semantic units defined outside of PREMIS.

Rationale	There may be a need to replace or extend PREMIS defined units. The extension is meant to be used as an alternative to the Statement, not to be used in combination with it.
Data constraint	Container
Repeatability	Repeatable
Obligation	Optional
Usage notes	For more granularity or to use externally defined semantic units, extensibility is provided. Either local semantic units or metadata using another specified metadata scheme may be included instead of or in addition to PREMIS defined semantic units. When using an extension schema, a reference to that schema must be provided. See further guidance in "Extensibility." Either *rightsStatement* or *rightsExtension* must be present if the Rights entity is included. If the *rightsExtension* container needs to be associated explicitly with any PREMIS subunit under *Rights*, the container *Rights* is repeated. Also, if extensions from different external schemas are needed, *Rights* should be repeated. It is recommended to give information about the metadata used in *rightsExtension* including date the metadata was created, status of the metadata, internal linking IDs, type of metadata used and its version, message digest and message digest algorithm of the metadata, and type of identifier for external metadata links.

Semantic unit	**4.4 linkingAgentIdentifier**
Semantic components	4.4.1 linkingAgentIdentifierType 4.4.2 linkingAgentIdentifierValue 4.4.3 linkingAgentRole
Definition	Identification of one or more Agents associated with the Rights statement.
Rationale	The Agent refers to the rightsholder that has either granted permission or required a restriction.. Rights statements may be associated with related Agents, either by linking from the Rights statement to the Agent(s) or by linking from the Agents(s) to the Rights statement. This provides the mechanism for the link from the Rights statement to the Agent.
Data constraint	Container

Repeatability	Repeatable
Obligation	Optional
Usage notes	Linking semantic units are mandatory in the sense that a repository needs to know the information, but are defined as optional because PREMIS does not specify in which direction the linkage should be. In particular, *linkingAgentIdentifier* is optional because a relevant Agent may be unknown, or no Agent may be relevant. The latter is likely when the Rights basis is statute.

Semantic unit	**4.4.1 linkingAgentIdentifierType**
Semantic components	None
Definition	A designation of the domain in which the linking Agent identifier is unique.
Data constraint	Must be an existing *agentIdentifierType* value.
Examples	[see examples for *agentIdentifierType*]
Repeatability	Not repeatable
Obligation	Mandatory

Semantic unit	4.4.2 linkingAgentIdentifierValue
Semantic components	None
Definition	The value of the *linkingAgentIdentifier*.
Data constraint	Must be an existing *agentIdentifierValue* value.
Examples	[see examples for *agentIdentifierValue*]
Repeatability	Not repeatable
Obligation	Mandatory

Semantic unit	4.4.3 linkingAgentRole
Semantic components	None
Definition	The role of the Agent in relation to the Rights statement.
Data constraint	Values should be taken from a controlled vocabulary.
Examples	contact creator publisher rightsholder grantor
Repeatability	Repeatable
Obligation	Optional

Semantic unit	4.5 linkingRightsIdentifier
Semantic components	4.5.1 linkingRightsIdentifierType 4.5.2 linkingRightsIdentifierValue 4.5.3 linkingRightsRole
Definition	An identifier for another Rights entity that is associated with this Rights entity, to be bundled such as to track Rights status over time.

Rationale	In previous iterations of PREMIS, there was no way to keep a bundle of multiple RightsStatements (R) together, such as stacks of previous iterations that must be tracked but are not current, or that are dependent on one another or trigger conditions. The linkingRightsIdentifier points to each applicable rightsEntity. The rightsEntity may have more than one rightsStatement. This linking between rightsStatements can also be applied in a situation where there is a dependency. For instance to reference rightStatements that can overrule a current rightsStatement. For example, under copyright the permission to provide access to the objects will be restricted by the copyright legislation. A license may lift the restrictions set by copyright. A second rightsStatement is needed. This statement describes the rights granted by the rightsholder (possibly the same Agent as is linked to the copyright-statement). In this case the license statement has priority over the copyright statement. This may also be used to track changing rights related to trigger situations. For example, the initial rightsStatement is based on contractual agreement with a publisher; a second rightsStatement based on a contractual agreement with publisher goes into effect when trigger conditions are met. It allows the repository to assert its amended rights when the condition is met: the second rightsStatement overrules the first and the history of the rights over time has been documented.		
Data constraint	Container		
Object category	**Representation**	**File**	**Bitstream**
Applicability	Applicable	Applicable	Applicable
Repeatability	Repeatable	Repeatable	Repeatable
Obligation	Optional	Optional	Optional
Usage notes	Linking semantic units are mandatory in the sense that a repository needs to know the information, but are defined as optional because PREMIS does not specify in which direction the linkage should be.		

Semantic unit	**4.5.1 linkingRightsIdentifierType**
Semantic components	None
Definition	A designation of the domain within which the *linkingRightsIdentifier* is unique.

Data constraint	Must be an existing *rightsIdentifierType* value.		
Object category	**Representation**	**File**	**Bitstream**
Applicability	Applicable	Applicable	Applicable
Examples		URI LCCN	
Repeatability	Not repeatable	Not repeatable	Not repeatable
Obligation	Mandatory	Mandatory	Mandatory

Semantic unit	**4.5.2 linkingRightsIdentifierValue**		
Semantic components	None		
Definition	The value of the *linkingRightsIdentifier*.		
Data constraint	Must be an existing *rightsIdentifierValue* value.		
Object category	**Representation**	**File**	**Bitstream**
Applicability	Applicable	Applicable	Applicable
Repeatability	Not repeatable	Not repeatable	Not repeatable
Obligation	Mandatory	Mandatory	Mandatory

Semantic unit	**4.5.3 linkingRightsRole**
Semantic components	None
Definition	The role of the linked Rights statement.
Data constraint	Values should be taken from a controlled vocabulary.
Examples	

Repeatability	Repeatable
Obligation	Optional

Glossary

Rights: Assertions of one or more legal entitlements, ~~or~~ permissions or obligations pertaining to a Digital Object and/or an Agent